

Part (b): Co-varying hyperparameter pairs (d8)

Compute-matched: d8, 614M tokens. 2D grids around the default recipe; on-axis points reuse part (a) runs. 18 new runs (off-axis combinations). AdamW (0.9, 0.95), linear decay, grad clip 1.0, data seed 42, single model seed.

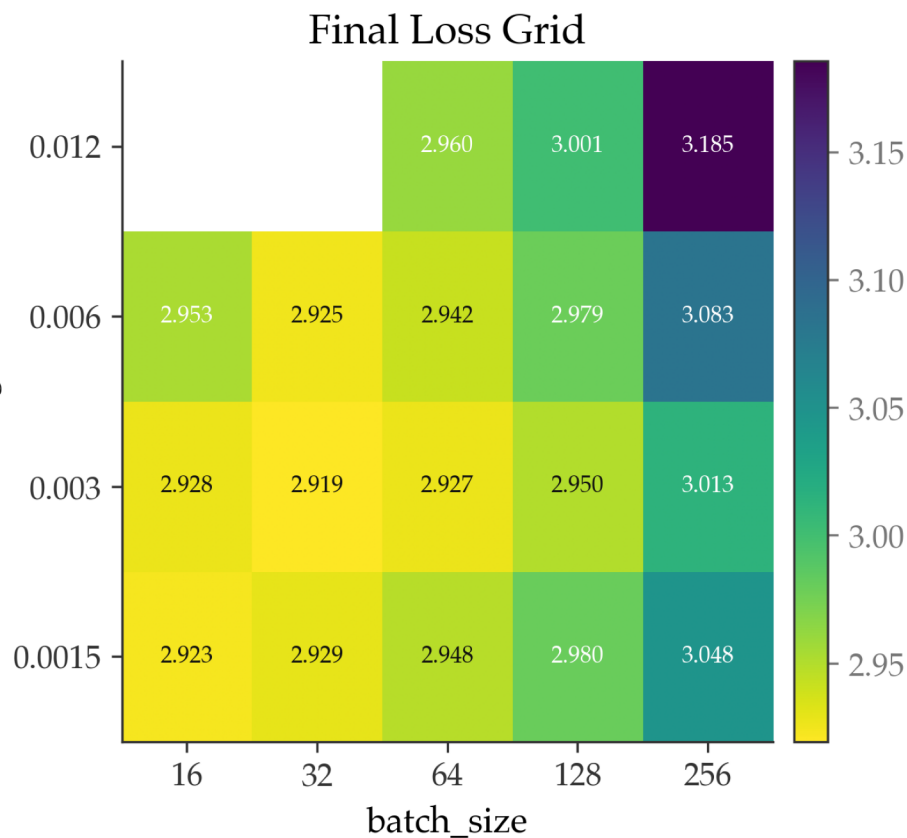
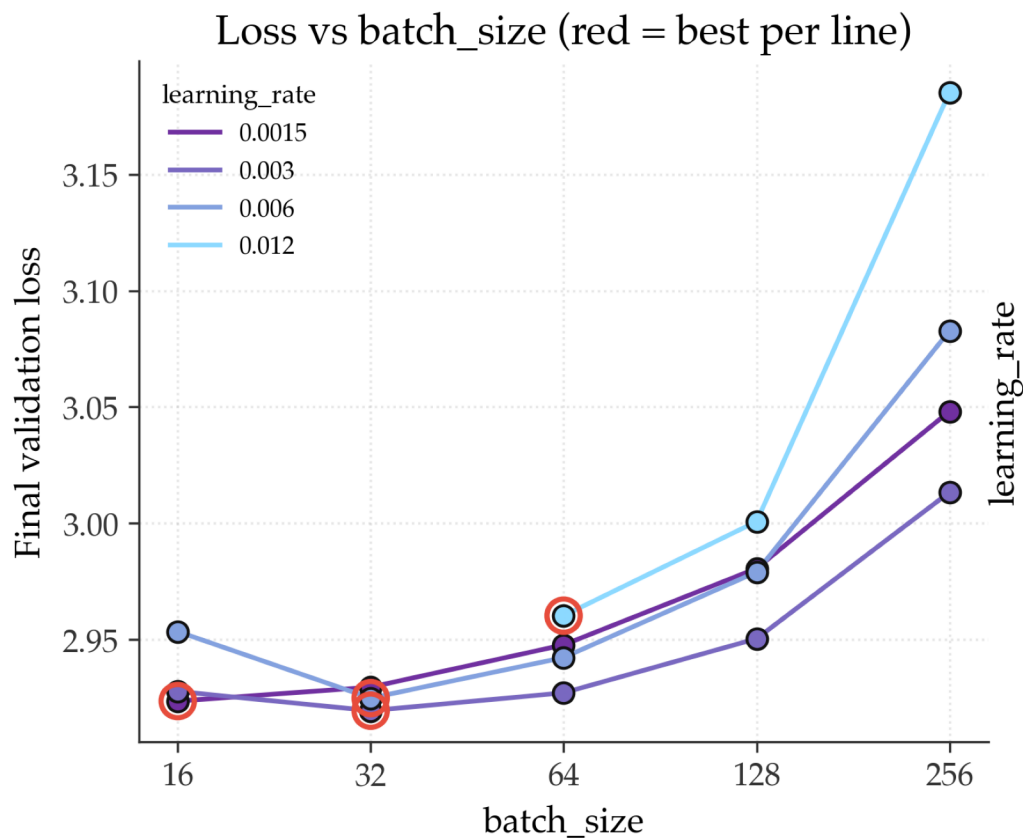
Pair	Grid	Interaction	Finding
Batch size x LR	16-256 x 1.5e-3, 3e-3, 6e-3 (+1.2e-2 for bs>=64)	Weak/partial	bs16 wants lower LR; bs >= 32 all best at 3e-3
LR x Weight decay	1.5e-3, 3e-3, 6e-3 x 0.05, 0.1, 0.2	Weak	Optima don't move; wd 0.2 helps more at low LR
LR x Warmup	1.5e-3, 3e-3, 6e-3 x 0.0025, 0.01, 0.04	Strong	Longer warmup -> higher best LR (6e-3 @ 4%)

Tune together: LR with warmup (and batch size at the small end). Mostly independent: weight decay.

Best run in (b): LR 6e-3, warmup 0.04, bs 64, wd 0.1 -> 2.916 (default recipe: 2.927).

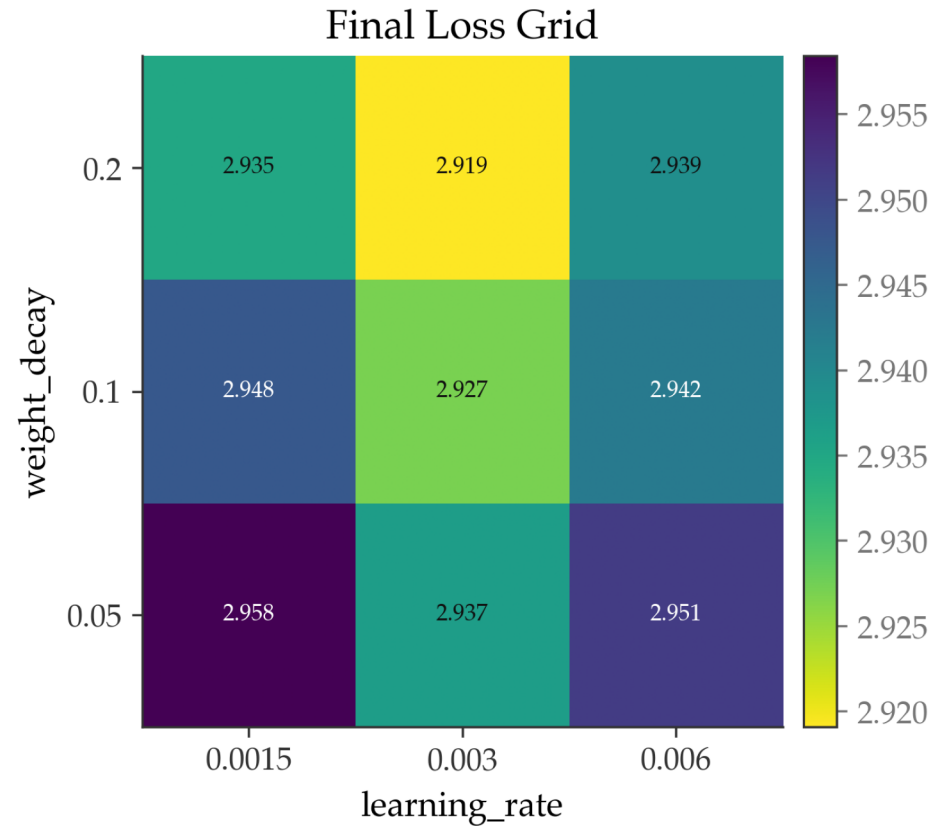
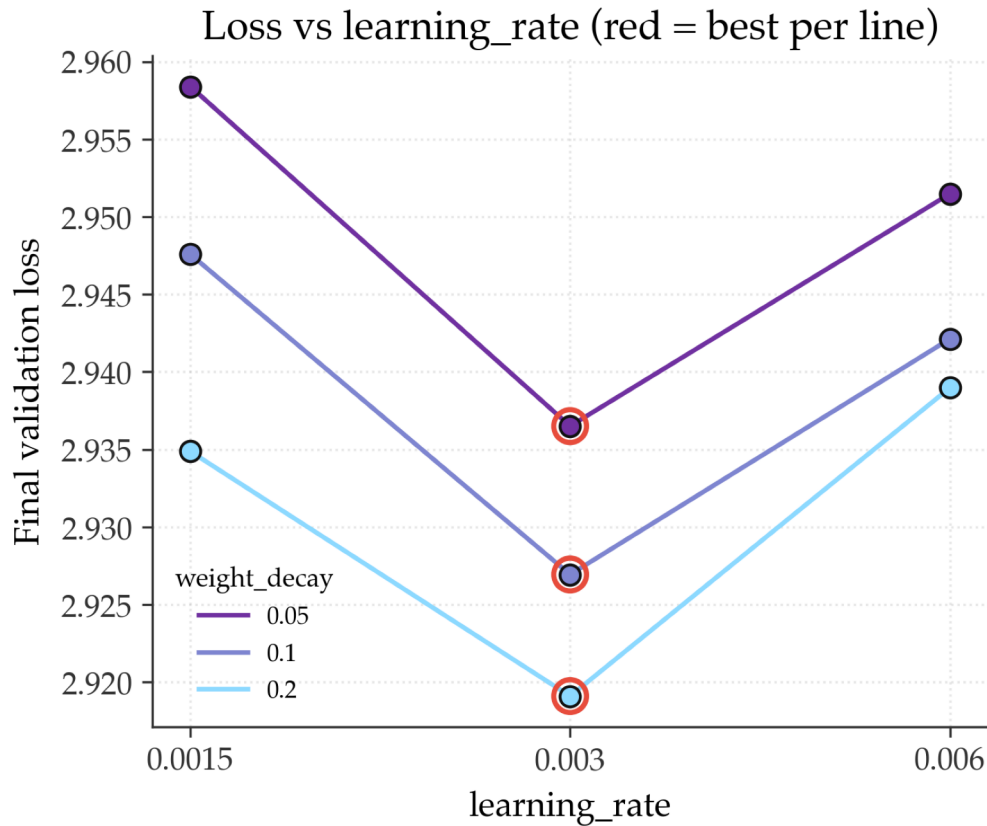
Note: differences below ~0.005 may be single-seed noise. One W&B run (f6zxa4r0) was relabeled after a logging mix-up; its data is LR 1.5e-3, wd 0.05. bs256 runs were rerun after an out-of-memory crash from Modal container reuse.

Batch size x Learning rate



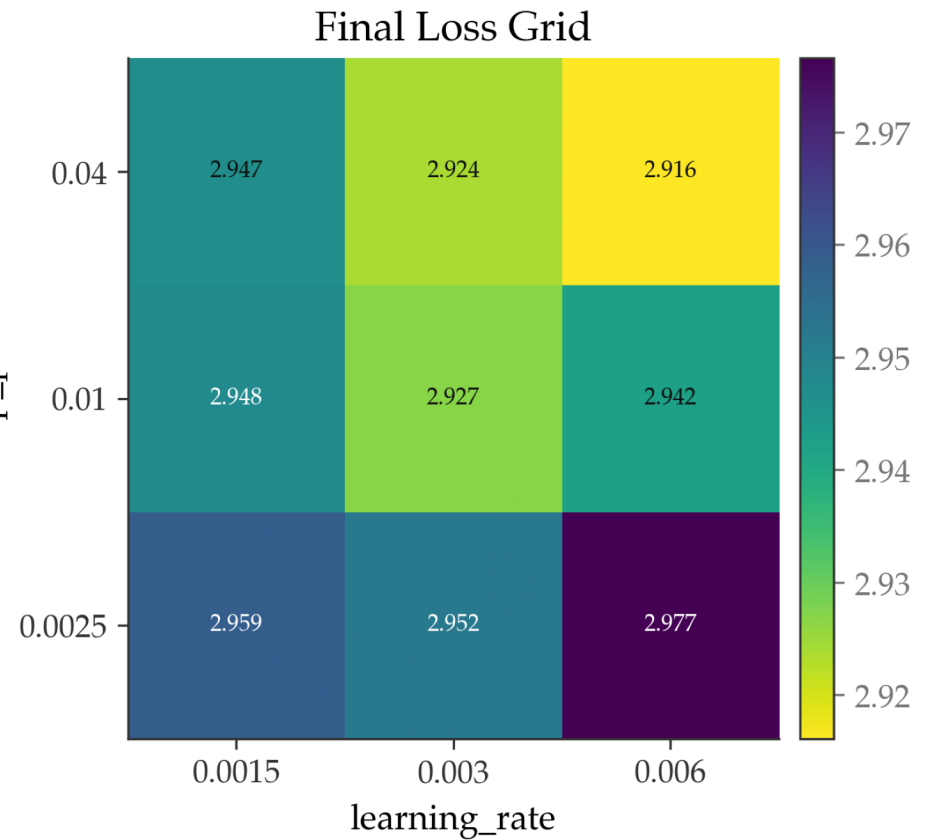
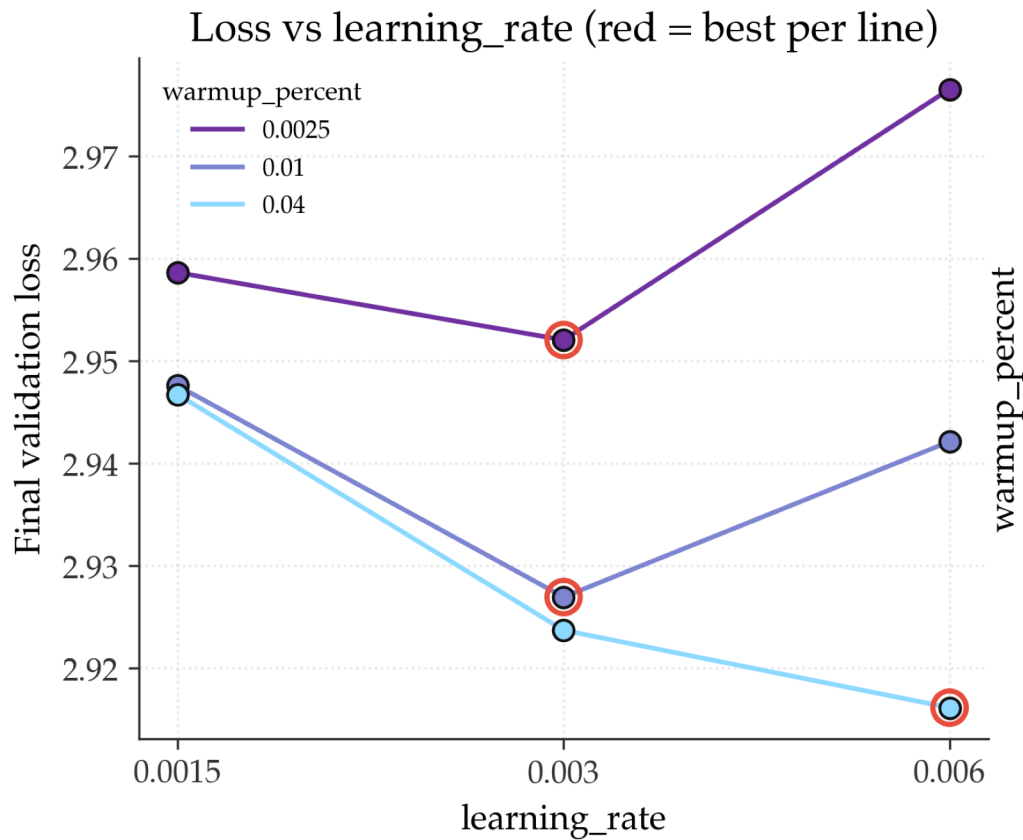
Best LR per batch size: 16 -> 1.5e-3; 32, 64, 128, 256 -> 3e-3. High LR hurts more as batch size grows (bs256: 1.2e-2 costs +0.17).
Hypothesis 'larger batch wants higher LR' NOT supported here; only the small-batch end shifts (bs16 prefers lower LR).

Learning rate x Weight decay



LR $3e-3$ is best at every weight decay; wd 0.2 is best at every LR. Weak interaction: wd 0.2 helps more at low LR (-0.013) than high LR (-0.003). Hypothesis 'best wd falls as LR rises' only weakly supported; the two can be tuned mostly independently.

Learning rate x Warmup



Strongest interaction: with 4% warmup the best LR moves up to 6e-3 (2.916, best run in (b)). Warmup's effect grows with LR: loss spread 0.012 at LR 1.5e-3 vs 0.061 at LR 6e-3. Hypothesis 'higher LR needs more warmup' supported.